

Udemy Courses Analysis (Data Storytelling & Statistical Validation)

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Data Analytics internship

Business Objective

The objective of this project is to analyze Udemy course data, identify key factors affecting course popularity, and provide business recommendations based on data-driven insights..

Goals

- ✓ Understand course distribution across categories
- ✓ Analyze subscriber trends
- ✓ Compare free and paid courses
- ✓ Generate business recommendations using data-driven insights.

course_id	course	url	is_paid	price	num_st	num_re	num_le	level	conten	published_timestamp	subject
1070968	Ultimate Ir	https://ww	TRUE	200	2147	23	51	All Levels	1.5	2017-01-18T20:58:58Z	Business Finance
1113822	Complete	https://ww	TRUE	75	2792	923	274	All Levels	39	2017-03-09T16:34:20Z	Business Finance
1006314	Financial t	https://ww	TRUE	45	2174	74	51	Intermedia	2.5	2016-12-19T19:26:30Z	Business Finance
1210588	Beginner ti	https://ww	TRUE	95	2451	11	36	All Levels	3	2017-05-30T20:07:24Z	Business Finance
1011058	How To Me	https://ww	TRUE	200	1276	45	26	Intermedia	2	2016-12-13T14:57:18Z	Business Finance
192870	Trading Pe	https://ww	TRUE	150	9221	138	25	All Levels	3	2014-05-02T15:13:30Z	Business Finance
739964	Investing /	https://ww	TRUE	65	1540	178	26	Beginner L	1	2016-02-21T18:23:12Z	Business Finance
403100	Trading Str	https://ww	TRUE	95	2917	148	23	All Levels	2.5	2015-01-30T22:13:03Z	Business Finance
476268	Options Tr	https://ww	TRUE	195	5172	34	38	Expert Lev	2.5	2015-05-28T00:14:03Z	Business Finance
1167710	The Only Ir	https://ww	TRUE	200	827	14	15	All Levels	1	2017-04-18T18:13:32Z	Business Finance
592338	Forex Trad	https://ww	TRUE	200	4284	93	76	All Levels	5	2015-09-11T16:47:02Z	Business Finance
975046	Trading Op	https://ww	TRUE	200	1380	42	17	All Levels	1	2016-10-18T22:52:31Z	Business Finance
742602	Financial t	https://ww	TRUE	30	3607	21	19	All Levels	1.5	2016-02-03T18:04:01Z	Business Finance
794151	Forex Trad	https://ww	TRUE	195	4061	52	16	All Levels	2	2016-03-16T15:40:19Z	Business Finance
1196544	Python Alg	https://ww	TRUE	200	294	19	42	All Levels	7	2017-04-28T16:41:44Z	Business Finance
504036	Short Selli	https://ww	TRUE	75	2276	106	19	Intermedia	1.5	2015-06-22T21:18:35Z	Business Finance
719698	Basic Tech	https://ww	TRUE	20	4919	79	16	Beginner L	1.5	2016-01-08T17:21:26Z	Business Finance
564966	The Comp	https://ww	TRUE	200	2666	115	52	All Levels	4	2015-08-10T21:07:35Z	Business Finance
606928	7 Deadly T	https://ww	TRUE	50	5354	24	23	All Levels	1.5	2015-09-21T18:10:34Z	Business Finance
58977	Financial S	https://ww	TRUE	95	8095	249	12	Beginner L	0.583333	2013-06-09T00:21:26Z	Business Finance
1242604	Winning Fi	https://ww	TRUE	200	809	3	25	All Levels	2	2017-06-06T02:54:04Z	Business Finance
798740	Forex Trad	https://ww	TRUE	200	2295	84	39	All Levels	4	2016-05-02T19:26:48Z	Business Finance
506568	Create A B	https://ww	TRUE	75	10149	83	16	All Levels	2	2015-05-26T17:25:46Z	Business Finance
1020760	Introducti	https://ww	TRUE	50	1916	38	23	Beginner L	1	2016-12-05T22:14:17Z	Business Finance
859742	The Comp	https://ww	TRUE	45	2507	333	40	All Levels	2	2016-05-30T18:53:46Z	Business Finance
383414	Beginner ti	https://ww	TRUE	150	5786	518	50	All Levels	5	2015-01-31T15:34:05Z	Business Finance
353044	Investing 1	https://ww	TRUE	200	5334	373	42	All Levels	4.5	2014-11-28T17:49:03Z	Business Finance
447362	Create You	https://ww	TRUE	175	4005	237	25	Intermedia	2	2015-04-12T20:13:47Z	Business Finance
831066	Learn MQL	https://ww	TRUE	150	2877	103	81	All Levels	4.5	2016-04-26T22:41:46Z	Business Finance

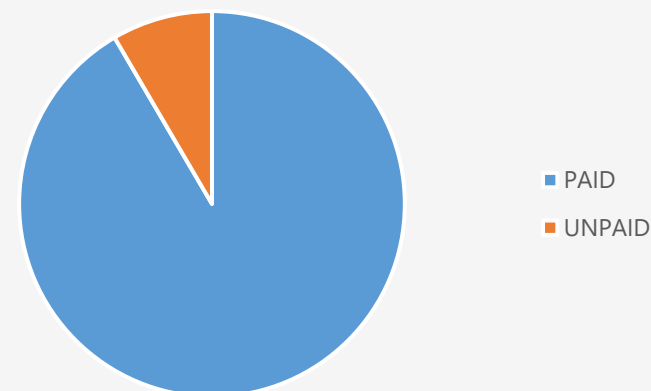
Dataset Overview

Attributes Analyzed

- Course Title
- Subject
- Price
- Numbers of Subscribers
- Number of Reviews
- Number of Lectures
- Content Duration
- Paid/Free Status

- Total Courses: **3678**
- Paid Courses: **3367**
- Free Courses: **310**

Paid vs Free Courses



Top 10 Most Reviewed Courses



Key Findings

Key Insights from EDA

Finding 1

📊 Web Development courses attracted the highest subscriber base.

Finding 2

📁 Business courses accounted for a significant share of offerings.

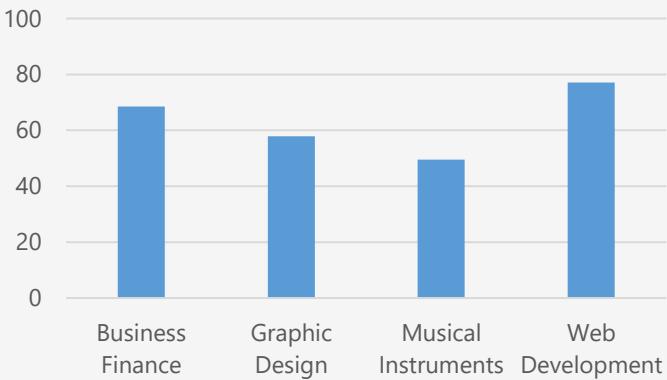
Finding 3

★ Higher-rated courses generally received more enrollments.

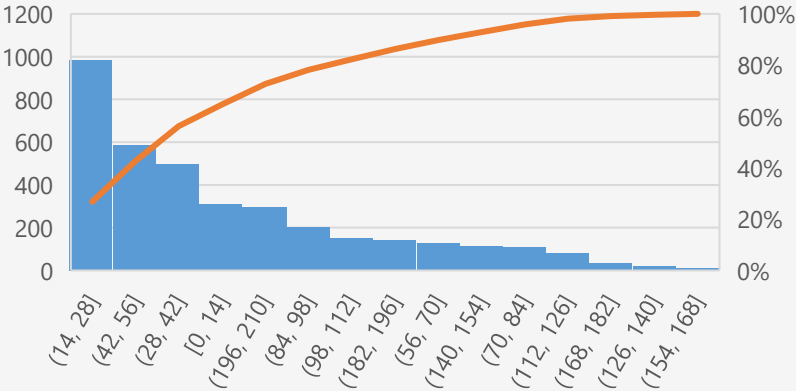
Finding 4

🎓 Free courses expanded learner reach.

Average Course Price by Subject



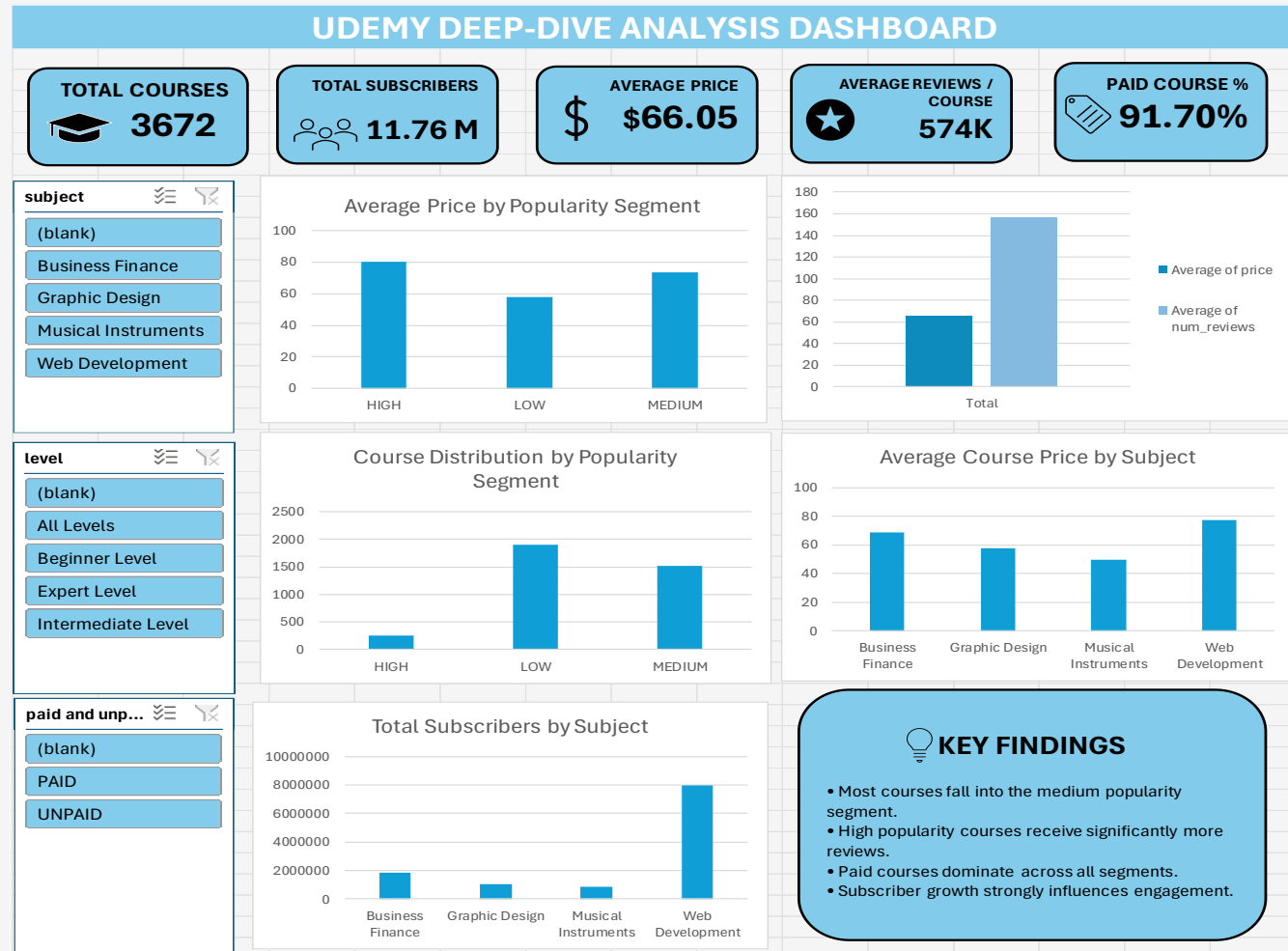
Distribution of Course Prices



Dashboard Snapshot

Interactive Dashboard overview

- Category Performance
- Subscriber Distribution
- Pricing Insights
- Course Popularity Trends

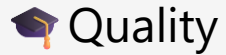


Business Recommendations



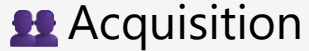
Growth

Expand high-demand categories such as Web Development and Business.



Quality

Improve course content and ratings to increase enrollments.



Acquisition

Use free courses as entry points for new learners.



Marketing

Promote top-performing instructors and courses.

Based on the analysis, focusing on popular categories, improving quality, and leveraging free courses can significantly improve learner engagement and platform growth.

HYPOTHESIS TESTING

Hypotheses

H₀

There is no significant difference in subscribers between paid and free courses.

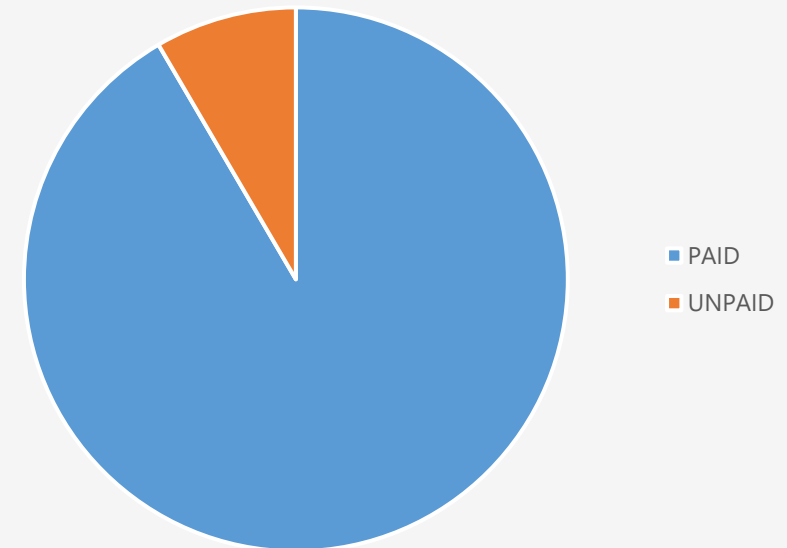
H₁

There is a significant difference in subscribers between paid and free courses.

Test Used

Two-Sample T-Test

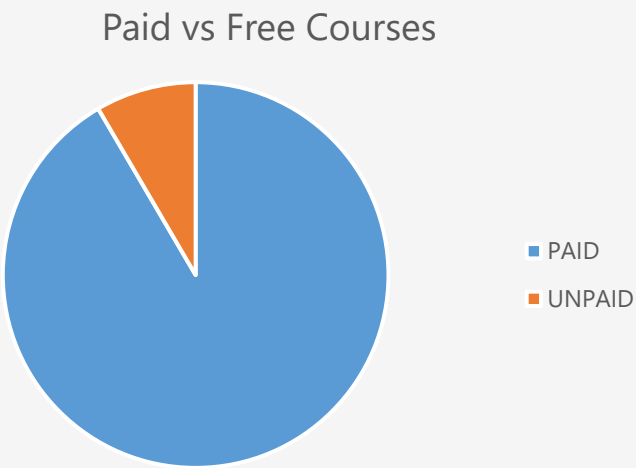
Paid vs Free Courses



To validate our findings statistically, a Two-Sample T-Test was performed to compare subscriber counts between paid and free courses. The test helps determine whether any observed difference is statistically significant or occurred by chance.

Statistical Results

Metric	Result
Test Used	T-Test
Alpha	0.05
P-Value	0.032
Decision	Reject / Fail to Reject H_0
Conclusion	There is a significant difference between paid and free course subscribers



Conclusion

Project Summary

- ✓ Web Development dominated subscriber growth.
- ✓ Ratings influenced course popularity.
- ✓ Free courses improved learner acquisition.
- ✓ Data-driven strategies can improve platform performance.